



Derivatives of Inverse Trigonometric Functions

Calculus Worksheet · Grade 11-12

Name: _____

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Learning Objectives

- Apply derivative formulas for inverse trigonometric functions
- Use the chain rule with inverse trig functions
- Combine product rule and inverse trig derivatives to differentiate composite expressions

Find the derivative of each function using the inverse trigonometric derivative formulas, simplifying your answer completely.

1. Find the derivative of the function.

$$y = \sqrt{\tan^{-1} x}$$

Answer: _____

2. Find the derivative of the function.

$$f(x) = \sin^{-1}(5x)$$

Answer: _____

3. Find the derivative of the function using the product rule.

$$f(x) = \sqrt{x^2 - 1} \cdot \sec^{-1} x$$

Answer: _____

4. Find the derivative of the function.

$$y = \cos^{-1}(3x^2)$$

Answer: _____

5. Find the derivative of the function.

$$y = \tan^{-1}(x^3)$$

Answer: _____

6. Find the derivative of the function.

$$f(x) = x \sin^{-1} x$$

Answer: _____



7. Find the derivative of the function.

$$y = \cot^{-1}(2x)$$

Answer: _____

8. Find the derivative of the function.

$$y = \sqrt{\sin^{-1} x}$$

Answer: _____

9. Find the derivative of the function.

$$f(x) = \tan^{-1}(e^x)$$

Answer: _____

10. Find the derivative of the function.

$$y = \csc^{-1}(x^2)$$

Answer: _____





Remind students to rewrite radicals as fractional exponents before differentiating and to memorize the six inverse trig derivative formulas.

Solutions

1. Find the derivative of the function.

$$y = \sqrt{\tan^{-1} x}$$

→ Rewrite the square root as the one-half power of inverse tangent of x .

→ Apply the chain rule with u equal to inverse tangent of x .

→ Multiply one-half times u to the negative one-half by the derivative of inverse tangent of x , which is one over one plus x squared.

→ Move the negative exponent to the denominator and simplify.

Answer:
$$y' = \frac{1}{2(1+x^2)\sqrt{\tan^{-1} x}}$$

2. Find the derivative of the function.

$$f(x) = \sin^{-1}(5x)$$

→ Identify u as $5x$ so that du/dx equals 5.

→ Apply the inverse sine derivative formula: one over the square root of one minus u squared.

→ Substitute $5x$ for u and multiply by the derivative of u , which is 5.

→ Simplify the squared term in the radicand to get $25x$ squared.

Answer:
$$f'(x) = \frac{5}{\sqrt{1-25x^2}}$$

3. Find the derivative of the function using the product rule.

$$f(x) = \sqrt{x^2 - 1} \cdot \sec^{-1} x$$

→ Rewrite the radical as x squared minus one to the one-half power.

→ Apply the product rule to the two factors.

→ Differentiate x squared minus one to the one-half using the chain rule.

→ Differentiate inverse secant of x using its formula and combine the two terms.

Answer:
$$f'(x) = \frac{x \sec^{-1} x}{\sqrt{x^2 - 1}} + \frac{\sqrt{x^2 - 1}}{|x|\sqrt{x^2 - 1}}$$

4. Find the derivative of the function.

$$y = \cos^{-1}(3x^2)$$

→ Let u equal $3x$ squared so that du/dx equals $6x$.

→ Apply the inverse cosine formula: negative one over the square root of one minus u squared.

→ Substitute u into the formula and multiply by du/dx .

→ Simplify the radicand to one minus $9x$ to the fourth.

Answer:
$$y' = \frac{-6x}{\sqrt{1-9x^4}}$$



5. Find the derivative of the function.

$$y = \tan^{-1}(x^3)$$

- Let u equal x cubed so that du/dx equals $3x$ squared.
- Apply the inverse tangent formula: one over one plus u squared.
- Substitute u and multiply by du/dx .
- Simplify the denominator to one plus x to the sixth.

Answer: $y' = \frac{3x^2}{1+x^6}$

6. Find the derivative of the function.

$$f(x) = x \sin^{-1} x$$

- Apply the product rule to x times inverse sine of x .
- Differentiate x to get 1 and keep inverse sine of x .
- Differentiate inverse sine of x using its formula.
- Combine the two terms to get the final derivative.

Answer: $f'(x) = \sin^{-1} x + \frac{x}{\sqrt{1-x^2}}$

7. Find the derivative of the function.

$$y = \cot^{-1}(2x)$$

- Let u equal $2x$ so that du/dx equals 2.
- Apply the inverse cotangent formula: negative one over one plus u squared.
- Substitute u and multiply by du/dx .
- Simplify the denominator to one plus $4x$ squared.

Answer: $y' = \frac{-2}{1+4x^2}$

8. Find the derivative of the function.

$$y = \sqrt{\sin^{-1} x}$$

- Rewrite the square root as inverse sine of x to the one-half power.
- Apply the chain rule with u equal to inverse sine of x .
- Multiply one-half times u to the negative one-half by the derivative of inverse sine.
- Move the negative exponent to the denominator and simplify.

Answer: $y' = \frac{1}{2\sqrt{\sin^{-1} x} \cdot \sqrt{1-x^2}}$

9. Find the derivative of the function.

$$f(x) = \tan^{-1}(e^x)$$

- Let u equal e to the x so that du/dx equals e to the x .
- Apply the inverse tangent formula: one over one plus u squared.
- Substitute u and multiply by du/dx .
- Simplify the denominator to one plus e to the $2x$.

Answer: $f'(x) = \frac{e^x}{1+e^{2x}}$



10. Find the derivative of the function.

$$y = \csc^{-1}(x^2)$$

→ Let u equal x squared so that du/dx equals $2x$.

→ Apply the inverse cosecant formula: negative one over the absolute value of u times the square root of u squared minus one.

→ Substitute u and multiply by du/dx .

→ Simplify by canceling an x to obtain the final answer.

Answer: $y' = \frac{-2}{|x|\sqrt{x^4 - 1}}$

